Module 2: Meter Reading & Water-Usage Input

HOUSEHOLD WATER-USAGE & BILLING MONITOR

Recording meter readings and calculating household water consumption



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Objective of Module 2

- Record the household's previous meter reading.
- Record the current meter reading.
- Calculate the total water consumed.
- Validate the entered readings.
- Prevent invalid or incorrect meter values.
- Pass the consumption value to the billing module.

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Meter Reading

A meter reading represents the amount of water recorded by the household's water meter.

The system uses two readings:

- **Previous Reading** — reading from the previous billing period
- **Current Reading** — latest meter reading

The difference between these readings gives the household's water consumption.



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Consumption Calculation

Formula

Water Consumption = Current Reading – Previous Reading

Example:

- Previous Reading = 1250 L
- Current Reading = 1430 L

Therefore:

Water Consumption = 1430 – 1250

✅ Water Consumption = 180 L

This value will be used by Module 3 for calculating the water bill.

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Java Class Design

```
class MeterReading {
    private double previousReading;
    private double currentReading;

    public MeterReading(double previousReading,
                        double currentReading) {
        this.previousReading = previousReading;
        this.currentReading = currentReading;
    }

    public double calculateConsumption() {
        return currentReading - previousReading;
    }
}
```

Concepts demonstrated:

- Class
- Objects
- Private variables
- Constructor
- Method
- Encapsulation

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Meter Reading Workflow

1 Enter Previous Reading

2 Enter Current Reading

3 Validate Readings

4 Calculate Consumption

5 Display Water Usage

6 Send Data to Module 3

This creates a smooth connection between meter reading and billing.

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Input Validation

The system should make sure that the readings are valid.

Conditions:

✔️ Current reading should not be less than previous reading.

✔️ Readings should not be negative.

✔️ Only valid numeric values should be accepted.

Example:

Previous Reading = 1500 L

Current Reading = 1350 L

❌ Invalid!

Because the current reading cannot normally be lower than the previous reading.

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Exception Handling

Invalid user input can cause errors during program execution.

Java's exception handling can prevent the program from crashing.

```
try {
    double reading = scanner.nextDouble();
}
catch (Exception e) {
    System.out.println("Invalid input!");
}
```

Benefits:

- Prevents sudden program termination
- Handles incorrect input
- Makes the application more reliable
- Improves user experience

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Module 2 Output

After entering the readings, the system displays:

Household: ABC Residence

Previous Reading: 1250 L

Current Reading: 1430 L

Water Consumed: 180 L

This consumption value is then passed to Module 3: Water Bill Calculation.



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What did we achieve?

- ✅ Captured meter readings
- ✅ Calculated water consumption
- ✅ Applied input validation
- ✅ Used Java classes and methods
- ✅ Applied encapsulation
- ✅ Added exception handling
- ✅ Prepared consumption data for billing

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